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RECEIVED
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REVISED
28 February 2026

ACCEPTED FOR PUBLICATION
9 April 2026

PUBLISHED
4 May 2026

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PAPER

Customizing a water balance model to the hydro-climate conditions of the Choke Mountain watershed in Ethiopia for water potential assessment and economic valuation

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Keywords: water balance, water potential, economic valuation, Choke mountain

Abstract

Numerous research efforts have explored novel strategies to alleviate water scarcity in various regions of Ethiopia. In this context, growing attention has been directed toward assessing the economic value of multiple watersheds to inform sustainable water resource management. This study develops and customizes a water balance (WB) model tailored to the hydro-climatic characteristics of the Choke Mountain watershed (CMW) to evaluate its water potential and support economic valuation. By integrating multiple data sources, the customized model provides improved alignment with natural hydrological dynamics compared to conventional WB models. Over the past four decades, groundwater levels in mountainous areas have declined at a faster rate than in lowland regions. The model incorporates altitude-sensitive precipitation estimates, highlighting the critical role of highland areas as water sources. The total annual water from terrestrial sources within the CMW is estimated at 13.86 billion m^3, contributing significantly to downstream flows across borders. The annual economic value of CMW water resources is estimated at $133.4 billion. Model results also indicate that a 10 m decline in peak elevation of the CMW leads to a 0.04 mm d^{-1} reduction in precipitation, highlighting the region's vulnerability to topographic and climatic changes. Furthermore, the Grand Ethiopian Renaissance Dam (GERD) is estimated to lose over 1.386 billion m^3 of water annually due to this decline, corresponding to an economic loss of $0.5 billion per year. These findings emphasize the need for comprehensive, national-scale economic valuation frameworks to support effective and sustainable water resource management strategies.

1. Introduction

Earth System Science provides a comprehensive framework for understanding interactions among the atmosphere, hydrosphere, lithosphere, and biosphere, offering critical insights into climate variability and sustainable resource management. In the Upper Blue Nile Basin (UBNB), studies have shown that atmospheric moisture transport and land–surface hydrological processes jointly govern regional water availability and climate resilience (Abera *et al* 2017, Megbar *et al* 2019). These interactions are especially

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pronounced in mountainous watersheds such as the Choke Mountain watershed (CMW), where steep topography and intensive land-use pressures amplify hydrological variability (Belayneh *et al* 2019).

The CMW, a key headwater system of the UBNB, encompasses an extensive highland block in central Gojjam, Ethiopia. Field studies and hydrological analyses indicate that the watershed sustains over 59 perennial rivers and approximately 270 springs, contributing significantly to downstream flow regimes (Yalew *et al* 2016, Belayneh *et al* 2019). Seasonal wetlands and open water bodies further enhance hydrological functions, support biodiversity, and stabilize ecosystems (Belay *et al* 2013, Belayneh *et al* 2019). Beyond its ground and surface water significance, the CMW governed the orographic precipitation to the east and the wet Gojjam vicinity. It is a cool temperate Wurch climatic zone and the four-stepped mountainous formations, locally known as Arat Mekeraker, sustain traditional livelihoods and indigenous knowledge systems (Belayneh *et al* 2019). High-value ecosystem products, including Asta-based honey and butter, are closely linked to the watershed's ecological integrity and represent an important component of local economies. These natural and cultural assets have contributed to international recognition of the area, including its designation as a tourism-oriented development zone. Indeed, due to its ecological, hydrological, and socio-economic importance, the CMW remains underrepresented in integrated hydrological economic studies (Yalew *et al* 2016, Belayneh *et al* 2019).

Recognizing water as a multipurpose economic asset that supports production, human welfare, and environmental services provides the economic foundation for evaluating the value of water resources (Mohammad-Azari *et al* 2020, Jingchai 2023). Water's direct uses, such as residential supplies, industry, and agriculture, as well as its indirect or non-market benefits, such as lowering risks, historical significance, and environmental protection, can be captured by giving it monetary value (Mohammad *et al* 2020). By indicating its limited supplies, opportunity costs, and trade-offs between competing uses, economic valuation facilitates effective allocation and fosters sustainable management (Birol *et al* 2006, 2010, Al-Juaidi *et al* 2014, Ortiz-Partida *et al* 2023). Additionally, it offers an open framework for pricing, cost recovery, policy formulation, and investment decisions, guaranteeing the conservation, equitable distribution, and resilience of water resources against growing demands from economic development, population expansion, and climate change (Mancosu *et al* 2015, Ortiz-Partida *et al* 2023), while limited attention has been given to quantifying water resources in economic terms. In this regard, it needs effective models to estimate the economic value of water resources.

In complex highland environments like the CMW, a water balance (WB) model provides a quantitative framework for economic valuation of water resources by systematically estimating the availability, distribution, and variability of water within a watershed (Musa *et al* 2017, Malik *et al* 2021, Yifru *et al* 2021, Ortiz-Partida *et al* 2023). By integrating key hydrological components such as precipitation, evapotranspiration, runoff, and groundwater recharge, the model translates physical water flows into measurable water supply potentials (Megbar *et al* 2019). These outputs form a critical basis for assigning economic value to water across competing uses, supporting efficient allocation, pricing, and sustainable management of water under changing climatic and socio-economic conditions.

Owing to this, different scholars studied the WB using existing fundamental models. For instance, Abera *et al* (2017) and Megbar *et al* (2019) over the UBNB and for the Chemoga watershed by Tadie (2024). Studies by Belayneh *et al* (2019), Yalew *et al* (2016), and others have assessed ecological and conservation aspects of the water resources. However, the lack of integrated economic valuation of the atmospheric precipitation, surface water, and groundwater resources in different parts of the globe, particularly in the CMW, is due to the limited performance of the fundamental WB model. To address these gaps, a fundamental WB model was improved and customized to the CMW by considering atmospheric and hydrological parameters for estimating the CMW water resources potential for economic valuation. Water potential resources are also forecast using an artificial neural network (ANN) model for future scenarios. Surface and groundwater potential (GWP) is evaluated in centum cubic feet (CCF), and economic valuation uses average water bill data for urban and rural users, implemented through MATLAB and climate data operators (CDO) tools (Alexandra *et al* 2021, Alzraiee *et al* 2024, Buchwald *et al* 2024). Furthermore, this integrated approach highlights both the hydrological and economic significance of the CMW, supporting informed resource allocation, sustainable development, and transboundary water cooperation.

2. Methodology

2.1. Location of the study area

The CMW covers an area of 27 722.25 km^2, located between latitudes 9.70° and 11.20° and longitudes 37.00° and 38.50° (figure 1). Elevations within the sub-basin range from 667 to 4100 m above sea level

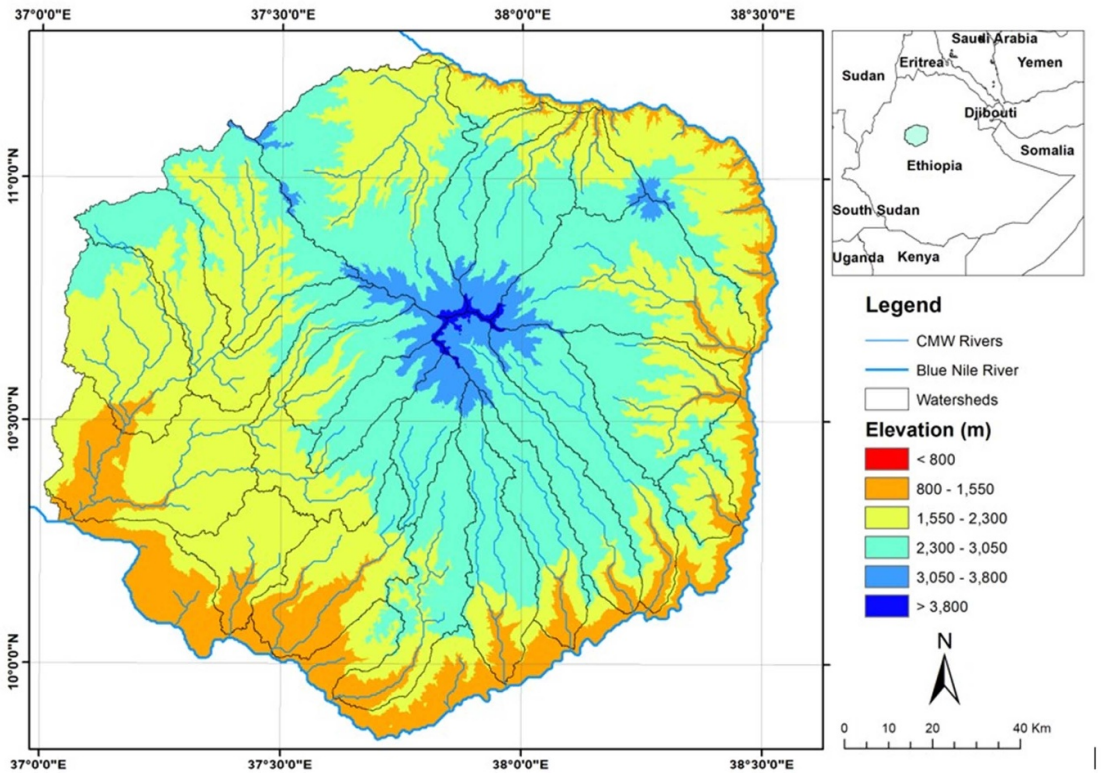


Figure 1. Location and topography of the CMW.

(Belay *et al* 2013). The region comprises approximately 5% Kolla, 27% Woyina Dega, 58.3% Dega, and 9.7% Woyina Dega zones. Annual temperature varies between 7.5 °C and 28 °C, while annual rainfall varies from 995 to 1,864 mm (Biratu *et al* 2022). As the headwaters of the UBNB, the CMW plays a vital role in the region's hydrology. Yalew *et al* (2016) noted that erosion from the highland areas in the central and eastern parts of the sub-basin is the primary source of sediment. The habitats mainly consist of moisture moorlands, sparsely populated with giant lobelias, grasses, and patches of woody plants (Biratu *et al* 2022). These ecosystems support many economic activities in the region and help maintain ecological balance. Therefore, it is recognized as a vital ecosystem at both national and international levels. To protect these resources, scientific research and education are essential to promote sustainable management.

2.2. Data types and sources

This study integrates atmospheric and groundwater availability using an interdisciplinary approach. Due to limited *in-situ* observations, a combination of satellite and reanalysis datasets, including ERA5 reanalysis, climate hazards infrared precipitation with station data (CHIRPS) (Aybar *et al* 2020), regional climate model (RCM), and Climate Research Unit (CRU), was investigated and analyzed. To validate the model and satellite data, precipitation (P) data from 20 *in-situ* stations operated by the national meteorology institute (1990–2024) were analyzed. The study also incorporates a wide range of hydro-climatic variables, such as evaporation (E) from soil, plants, and open water (excluding oceans), runoff (Ro), inflow and outflow of groundwater (G), large-scale and convective precipitation (P), air temperature (T), soil temperature (ST) across different layers (0 m, 7 m, 28 m, and 100 m depth), specific humidity (SH), relative humidity (q), 10 m zonal wind (U), 10 m meridional wind (V), pressure levels (Pl), potential evaporation (PE), vegetation cover (VC), thermal atmospheric energy (TE), soil water volumetric content (SWV) across various layers (0 m, 7 m, 28 m, and 100 m depth), soil type (S. Type), and flux data for E, P, and Ro sourced from ERA5 from 1990 to 2024. The temporal and spatial resolutions of all datasets are summarized in table 1.

2.3. Methods of data analysis

All datasets were interpolated and reprojected using Kriging methods implemented in the CDO, and harmonized to a uniform spatial resolution of 0.05° × 0.05° with daily temporal resolution. Gaps in *in situ* data were filled using weighted average interpolation (Abera *et al* 2017). Satellite and reanalysis datasets

Table 1. Data sources, spatial and temporal resolution, time coverage, and corresponding websites.

Data types	Data used	Time resolution	Space resolution	Website
In-situ	1990–2024	Daily	20 per CMW	Non
RCM	1990–2024	Daily	0.22 by 0.22°	cds.climate.copernicus.eu
CHIRPS	1990–2024	Daily	0.05 by 0.05°	www.CHIRPS-3.0/global_daily
ERA5	1990–2024	Daily	0.10 by 0.10°	cds.climate.copernicus.eu
CRU	1990–2024	Monthly	0.50 by 0.50°	www.cru.uea.ac.uk

were evaluated against *in-situ* observations using bias ratio (BR), root mean square error (RMSE), and Pearson correlation coefficient (R^2). ANN model Climate prediction by altitude class based on various scenarios was analyzed up to 2030 by splitting the data training (80%) and test (20%) using a feed-forward multilayer perceptron error metrics (BR, MRSE, MAE, and R^2) (Pregger *et al* 2020, Schweizer and Lazurko 2020, 2020, Sharma *et al* 2020). Decadal variations in VC, ST, and SWV across multiple soil depths were analyzed for the periods 1991–2000, 2001–2010, 2011–2020, and 2021–2030.

2.3.1. WB model formulation

The WB model was improved to incorporate soil moisture and atmospheric moisture budgets using continuity equations (Inflow (IF) and outflow (OF)) by following the equations,

$$\partial tV = \mathrm{IF} - \mathrm{OF}. \tag{1}$$

Hydraulic gradient (J) and groundwater layer depth (L) were estimated using a modified Bernoulli-based Lagrangian transport model (Sharma *et al* 2020)

$$J = \frac{\partial L}{\partial r}(r,t) + \frac{\partial^2 L}{\partial r^2}(r,t). \tag{2}$$

Inflow is given by:

$$\mathrm{IF} = k\left[\frac{\partial L}{\partial r}(r,t) + \frac{\partial^2 L}{\partial r^2}(r,t) * 2\pi\,(r+\mathrm{d}r) * b\right] \tag{3}$$

where k is the water loss coefficient, b is the thickness of the depth (L_n–$L_{\mathrm{n}-1}$), and r is the radial distance (0–83.25 km-the radius of the CMW).

OF and water storage (WS) follow:

$$\mathrm{OF} = k\left[\frac{\partial L}{\partial r}(r,t) * (2\pi rb)\right] \tag{4}$$

$$\partial V = 2\pi r(\mathrm{WS} * \mathrm{d}r * \mathrm{d}L). \tag{5}$$

In WB modeling, specific humidity is a critical input parameter for estimating the vertical flux of integrated soil water storage (Q). This flux is calculated using the following equation:

$$Q(x,y,t) = \frac{-1}{g}\left(\int_{Ps}^{Pd} \mathrm{SH}(x,y,t) * U(x,y,t)\right)\mathrm{d}P \tag{6}$$

where SH is specific humidity (gm gm^{-1}), U (m s^{-1}) is the motion of fluid in the soil in east-to-west and north-to-south components, p is pressure (Pa), g is the acceleration due to gravity (9.81 m s^{-2}), p_s is the pressure at the surface level (0), and p_d is the pressure at the depth level (100 m). The amount of change in water storage (WS) (based on the amount of water content of the soil or groundwater via pressure) is calculated by

$$\mathrm{WS} = \frac{-1}{g}\frac{\partial}{\partial t}\int_{Ps}^{Pd} Q\mathrm{d}P. \tag{7}$$

The change in storage in the improved WB model (IWB) was calculated by

$$\mathrm{IWB} = \frac{1}{A}(\mathrm{IF} - \mathrm{OF}). \tag{8}$$

The basic WB model (OWB) is calculated by

$$\mathrm{OWB} = P - (R + G + E). \tag{9}$$

In those equations, P represents precipitation, R is runoff, E denotes total evaporation, and G is groundwater flow (including base flow of rivers, mines, seepage from rivers into aquifers), all obtained from the ERA5 reanalysis datasets. This study focuses on comparing the improved and original models for the CMW, particularly in terms of soil moisture budget and water storage. The performance of the improved model is evaluated using the following metric:

$$\mathrm{ER} = \mathrm{IWB} - \mathrm{OWB} \tag{10}$$

where ER is the error between the basic (OWB) model and the improved (IWB) model. Hence, the volumetric soil GWP at different depth levels (L_1–L_4) ($\mathrm{SWVL}_{1\text{–}4}$) model is calculated by equation (11), which is the new application of the study area since the IWB is calculated by considering soil and atmospheric moisture budgets,

$$\mathrm{VSW} = J * \mathrm{WB} * \partial V - E \tag{11}$$

where E is total evaporation and is calculated as follows:

$$E = ET_{\mathrm{int}} + ET_{\mathrm{bsoil}} + ET_{\mathrm{ow}} + ET_{\mathrm{veg}} + ET_{\mathrm{imp}} + ET_{\mathrm{unst}} + ET_{\mathrm{gw}} \tag{12}$$

E accounts for the loss of water in several forms, including interception (ET_{int}), ET from bare soil (ET_{bSoil}), ET from open water (ET_{ow}), transpiration (ET_{veg}), ET from impervious areas (ET_{imp}), ET from unsaturated zones (ET_{unst}), and groundwater ET (ET_{gw}).

2.3.2. Atmospheric, Earth surface, and GWP modeling

Atmospheric precipitation (water potential) (P), because the Choke Mountain (CM) is represented by the model equation (13),

$$P = \mu * \Phi * w_r * P_\omega + \mathrm{CP} \tag{13}$$

where P is precipitation, μ is the unit adjustment constant, P_ω is precipitable water vapor, w_r is vertical wind shear, and CP is convective precipitation from the local moisture. In order to calculate the vertical windshear w_r equation (14) is a fundamental requirement,

$$\mathrm{RU} = Ui + Vj + wk \tag{14}$$

where RU is resultant wind, U is zonal wind, V is meridional wind, and w is vertical wind. The new model development is focusing on the role of mountains in converting advection wind to vertical wind shear. Therefore, find the k direction of the advection change to the convective scheme. The curl of the RU is

$$\Delta X\mathrm{RU} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right) i + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right) j + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) k. \tag{15}$$

The k-direction component is valuable to find the wind shear, hence

$$w_r = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) k. \tag{16}$$

To quantify the role of the CM for large-scale P contribution to the CMW, the geopotential height (GPH) represented by Φ is required and calculated by

$$\Phi = \int_{z=0}^{z=4094\mathrm{m}} g(z)\,\mathrm{d}z. \tag{17}$$

The altitude of CM is 4094 m, and the vertical integral component of the atmospheric moisture budget has a high indication of the occurrence of P. Owing to this, the precipitable water vapor is calculated by

$$P_\omega = \frac{1}{g}\frac{\partial}{\partial t}\int_0^{1000} q\mathrm{d}p \tag{18}$$

where q is relative humidity, dp is pressure from 0 to 1000 hpa. Equations (16)–(18) are the input values of the fundamental model, equation (13).

Then the annual average of 35 years of groundwater potential (GWP) is modeled by

$$\mathrm{GWP} = \int_{L=0}^{L=1\mathrm{m}} \mathrm{SWV}\mathrm{d}L \tag{19}$$

where GWP is groundwater potential, SWV is soil water volumetric content, and L is the soil layer depth at $L_1 = 0$ m, $L_2 = 7$ m, $L_3 = 28$ m, and $L_4 = 100$ m from the surface to the depth of the ground. However, the economic valuation of the water potential is calculated at the surface level. Hence, we take L from 0 to 1.

To calculate the GWP from only the ground sources (GWPS) during the dry season (December, January, and February), 35 years of data were outlined by removing the storage parts:

$$\mathrm{GWPS} = \mathrm{GWP} - \mathrm{WS} \tag{20}$$

where GWP is calculated from equation (19) only from dry seasons, and WS is calculated from equation (7) from dry seasons. The atmospheric and underground water potential is only from ground sources. To remove the two-way summation of the two components, the surface water is excluded, and it is modeled by

$$\mathrm{TWP} = P + \mathrm{GWPS}. \tag{21}$$

In this formulation, TWP denotes the total water potential contributed by both atmospheric inputs and subsurface sources. Soil water losses were expressed as equivalent water depth over the CM domain and computed using the following equation:

$$H = \frac{1}{A}\int_{L=0}^{L=100} \mathrm{GWP}\mathrm{d}L \tag{22}$$

where H is the depth, A is the unit area (1 m^2) per pixel point of the CMW. The model represented by equation (22) is quite useful for studying landslides in the study area. Finally, the CMW water potential economy value (CMWEV) is modeled by

$$\mathrm{CMWEV} = \mathrm{CCF}^{*}\mathrm{MAB} \tag{23}$$

where CCF is centum (hundred) cubic feet. 1CCF = 748 Gallons = 2.83 m^3 (Alzraiee *et al* 2024). The minimum value of the bill inside the city (MBIC) = \$14.3, and the minimum bill value outside the city (MBOC) = \$23.2 (Alexandra *et al* 2021, Buchwald *et al* 2024). In this paper, the average value of MBIC and MBOC (MAB) was considered for calculating the economic valuation of the CMW. This has been suggested by several scholars (Birol *et al* 2006, 2010, Al-Juaidi *et al* 2014, Ortiz-Partida *et al* 2023). Hence, the TWP value in m^3 is converted to CCF and substituted into equation (23), which enables the calculation of the economic value of the CMW in both atmospheric and surface water from underground resources. All modeling and analysis were performed using MATLAB and CDO tools.

3. Results

3.1. Validation of satellite and model precipitation data against in-situ observations

To evaluate the performance of satellite-based and model-derived P datasets over the CMW, *in-situ* observations were compared with four commonly used products: ERA5, CHIRPS, CRU, and a RCM. The analysis yielded correlation coefficients (R^2) of 0.85 for ERA5, 0.73 for CHIRPS, 0.81 for CRU, and 0.61 for the RCM (table 2). Corresponding bias ratios (BR) were 0.99, 1.11, 1.06, and 1.06, while mean relative errors (MRE) were −0.02, 0.12, 0.04, and 0.09, respectively. Root mean square errors (RMSE) were estimated at 0.61 (ERA5), 2.82 (CHIRPS), 0.73 (CRU), and 0.45 (RCM) (table 2). Overall, ERA5 demonstrated the strongest agreement with ground observations, indicating superior capability in capturing spatial and temporal P variability across the CMW. CHIRPS and CRU exhibited moderate performance, whereas the RCM showed comparatively weaker correlation, despite having the lowest RMSE and the second-lowest MRE after ERA5. Based on this comparative evaluation, ERA5 was selected as the primary P input for subsequent WB modeling. These results align with earlier studies in the UBNB, which demonstrated that ECMWF products outperform other datasets in reproducing observed precipitation patterns (Abera *et al* 2017, Megbar *et al* 2019).

Table 2. Error metrics calculated between reanalysis and gauge-based precipitation data.

Gauge	BR	MRE	RMSE	R^2
ERA5	0.99	−0.02	0.61	0.85
CHIRPS	1.11	0.12	2.82	0.73
RCM	1.06	0.04	0.73	0.81
CRU	1.06	0.09	0.45	0.61

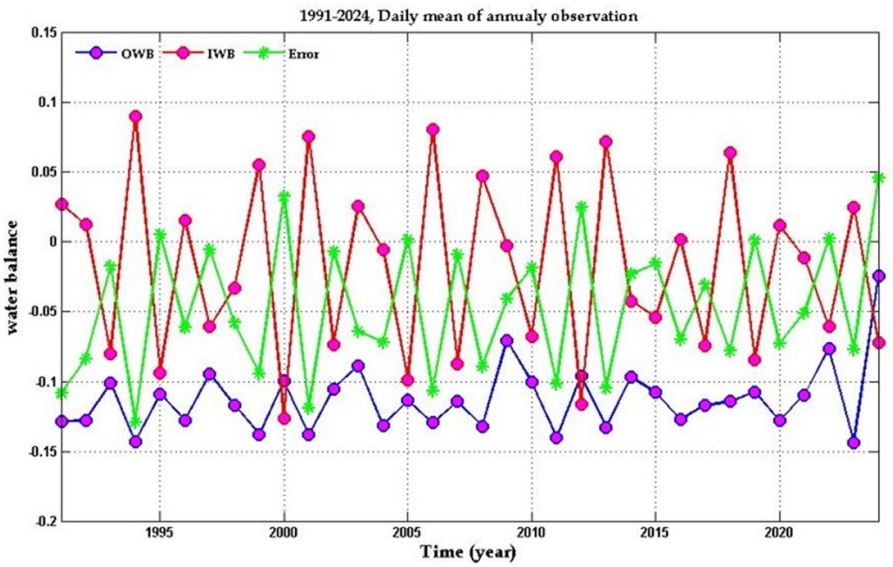


Figure 2. Temporal variability of OWB and IWB over CMW from 1991–2024.

3.2. Spatiotemporal variation of hydro-climate parameters over the CMW, Ethiopia

Figure 2 presents the outputs of the OWB and the IWB models applied to the CMW. The OWB model estimates ranged from −0.15 to −0.03 mm, while the IWB model varied from −0.10 to 0.09 mm. The average deviation between the two models is approximately—0.05 mm, suggesting that the IWB accounts for additional hydrological variability not captured by the OWB. Time series analysis from 1991 to 2024 reveals that the OWB model exhibits a weak downward trend, dominated by consistently negative (unidirectional) fluctuations. In contrast, the IWB model displays both positive and negative oscillations, reflecting more dynamic behavior that aligns with expected seasonal and interannual hydrological cycles. This contrast indicates that the OWB model fails to preserve mass continuity, while the IWB provides a more physically realistic representation of WB dynamics.

Further evidence for the enhanced performance of the IWB model comes from similar studies conducted in other Ethiopian watersheds. For example, Tadie Mulie (2024) applied the SWAT model in the Chemoga watershed and reported a gradual increase in WB over time, a trend that is consistent with our OWB results. However, unlike the OWB approach, the IWB model captures finer-scale variability, reflecting its enhanced sensitivity to spatial heterogeneity in topography and land-cover patterns. Similar WB trends were reported by Bedasa and Yilma (2024) in the Erer sub-basin and by Tadess and Jothimani (2024) in the Wabe River basin, further corroborating the robustness of the IWB framework. These comparisons highlight that while conventional OWB models are effective in representing basin-scale trends, the IWB approach provides a more realistic representation of hydrological dynamics by explicitly accounting for local physiographic and climatic controls, an advantage that is particularly critical in topographically complex CMW.

Spatial patterns of key hydroclimatic variables further validate the model improvements. E is lowest in the eastern part of the CMW, likely due to sparse vegetation and lower rainfall (figure 3 at panel (a), and figure 4 at panel (a)). Conversely, the central and northern areas, especially those surrounding CM, show elevated E values driven by dense forest cover, steep slopes, and orographic precipitation. Ro is also highest in the CM and Abbay Gorge areas, where steep terrain and reduced vegetation limit infiltration. P, Ro, and groundwater recharge (G) are concentrated around CM, demonstrating the critical influence of topography on spatial water dynamics (figure 3 panels (b) and (c)). The presence of negative E and

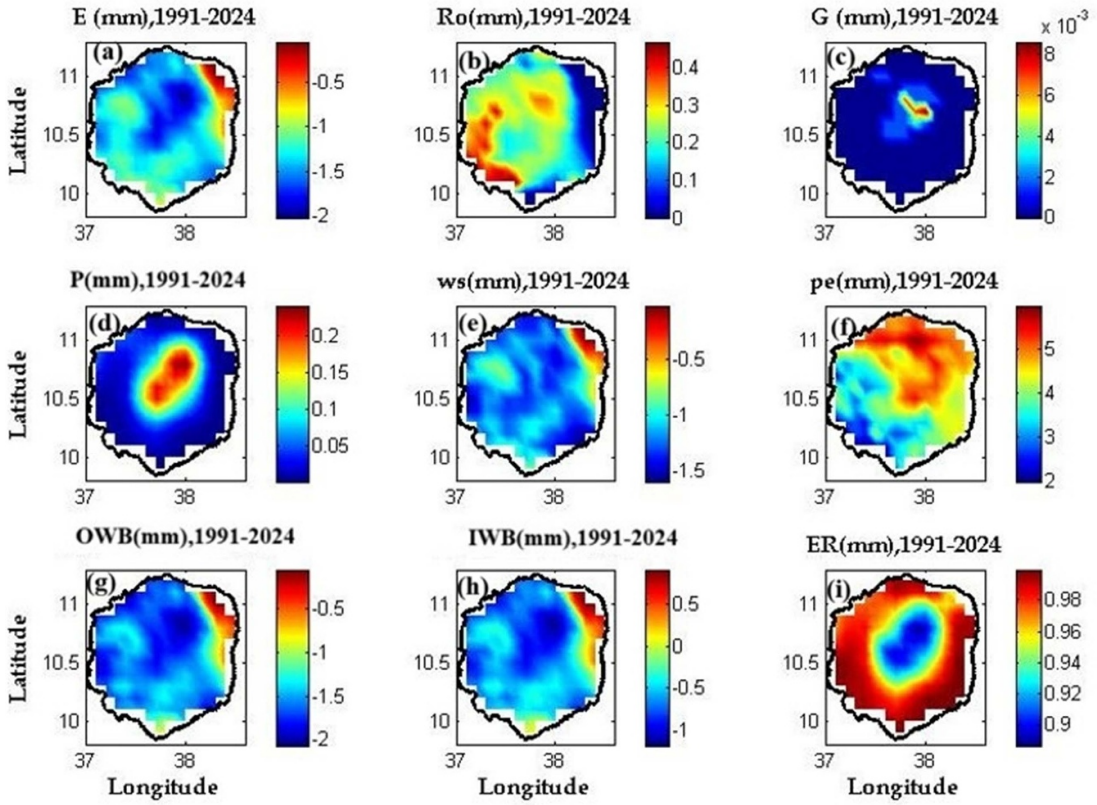


Figure 3. Spatial distribution and variation of total evapotranspiration (*E*, panel (a)), runoff (*Ro*, panel (b)), groundwater recharge (*G*, panel (c)), total precipitation (*P*, panel (d)), WS, panel (e)), effective precipitation (Pe, panel (f)), OWB, panel (g)), IWB, panel (h)), and error ratio (ER, panel (i)) over the CMW from 1991 to 2024.

WS values in some areas may reflect downward moisture movement (e.g. recharge zones) or indicate limitations in the model structure or resolution.

The northeastern region of the CMW displays the lowest WS values, likely due to land degradation and sparse vegetative cover, while the central, northern, and southeastern zones show relatively higher storage (figure 3 panel (e)). Despite receiving the highest *P*, the area around CM shows negative IWB values approaching −1 mm, while the eastern lowlands approach zero. This contrast highlights the modulating role of elevation, slope, and land cover on the spatial distribution of WB. Comparative analysis of the OWB and IWB outputs (figure 3, panels (g) and (h)) underscores the improvements achieved through model refinement. The OWB model consistently predicts negative net balances throughout the watershed, failing to account for local water accumulation or seasonal surplus. In contrast, the IWB model captures both surpluses and deficits, yielding a more balanced and physically consistent hydrological picture. Panel (i) of figure 3 clearly illustrates the spatial difference between the two models.

WS, derived from the atmospheric and surface moisture budget (figure 3, panel (e)), ranges from −1.5 to +0.5 mm (observed). The OWB model estimates span from −2.0 to −0.5 mm (mean: −1.2 mm), while the IWB model ranges from −1.0 to +0.5 mm (mean: −0.76 mm). The smaller deviation between WS and IWB (–0.21 mm), compared to WS and OWB (–0.23 mm), suggests that the IWB model more accurately represents the region's hydrological conditions. Importantly, the IWB model reveals spatial patterns of climate vulnerability, particularly identifying mountainous regions as more sensitive to climate-induced water stress. In contrast, the OWB model depicts uniformly negative balances across the basin, obscuring important spatial variations. This finding is consistent with Mulvaney *et al* (2022), who reported heightened climate sensitivity in highland areas compared to lowland zones. The relationships between the IWB model outputs and major hydroclimatic variables are summarized in table 3. Statistically, *E* and *Ro* show strong negative correlations with IWB ($R^2 = -0.60$, $p < 0.01$; $R^2 = -0.50$, $p = 0.05$), indicating that increased water losses through *E* and *Ro* reduce the net WB. WS and *P* exhibit strong positive correlations with IWB ($R^2 = +0.71$, $p = 0.02$; $R^2 = +0.58$, $p = 0.01$), confirming their critical role in sustaining water availability.

Table 3. Spatiotemporal average values of the R^2 and corresponding *P*-values for hydro-climatic parameters in the Choke Mountain watershed (CMW), Ethiopia.

Parameters	R^2	*P*-value
IWB and *P*	0.58	0.01
IWB and *T*	0.18	1.00
IWB and *E*	−0.60	0.00
IWB and WS	0.71	0.02
IWB and *G*	−0.14	0.19
IWB and *Ro*	−0.50	0.05

Table 4. The accuracy of the ANN model predictions was evaluated using several error metrics: BR, MRE, RMSE, R^2, and *P*-value for *P*, *T*, and ST. These were compared between observed and predicted values.

Type	BR	MRE	RMSE	R^2	*P*
P	0.99	−0.09	0.61	0.95	0.01
ST	1.06	0.04	0.73	0.96	0.01
T	1.11	0.12	2.82	0.92	0.02

3.3. ANN model prediction performance

Comparing observed and predicted values across the training, validation, test, and combined datasets revealed strong linear relationships (table 4). The performance of the ANN model was evaluated using BR, or MRE, RMSE, R^2, and statistical significance (*p*-value) for *P*, ST, and *T*. Overall, the model showed strong predictive skill for *P*, with BR $\approx$ 0.99, MRE $\approx$ −0.09 mm, RMSE $\approx$ 0.61 mm, and $R^2 = 0.95$, and $p = 0.01$. Similarly, ST predictions demonstrated very good agreement with observations (BR $\approx$ 1.06, MRE $\approx$ 0.04 K, RMSE $\approx$ 0.73 K, $R^2 = 0.96$, $p = 0.01$). *T* predictions, although statistically significant ($R^2 = 0.92$, $p = 0.02$), exhibited comparatively higher uncertainty, with a positive bias (BR $\approx$ 1.11, MRE $\approx$ 0.12 K) and an RMSE of 2.82 K (table 4). These results indicate that the ANN model accurately reproduces the magnitude and temporal variability of *P* occurrence, while showing moderate but noticeable overestimation in *T*. These high correlations are consistent with those reported in ANN model-based climate and hydrological forecasting applications (Goyal *et al* 2014). However, the fitted regression lines exhibited slopes slightly below unity (e.g. 0.84 for training and 0.87 for combined data), indicating a tendency of the model to underestimate high values and overestimate low values. For robust environmental modeling, additional metrics such as RMSE, MAE, and especially the Nash–Sutcliffe Efficiency (NSE) are recommended (Moriasi *et al* 2007, Kisi and Parmar 2016). Overall, the ANN model demonstrated a strong capacity to represent complex climatic relationships, as recommended by recent reviews in climate–AI modeling (Shamshirband *et al* 2020, Li *et al* 2024).

3.4. Spatial and decadal analysis of land surface and climate parameters

This section presents a decadal and spatial analysis of key biophysical variables: S. type, VC, thermal energy (TE), *P*, *T*, and average soil temperature (aST) across soil layers ($L_1 = 0$ m, $L_2 = 7$ m, $L_3 = 28$ m, and $L_4 = 100$ m) depth within the CMW.

The analysis spans four decades: The first decade (1990–2000), the second decade (2001–2010), the third decade (2011–2020), and the projected fourth decade (2021–2030). In the first decade, VC reached a maximum value of 0.7 in the CM and southwestern regions, with the minimum value (0) observed in the northern and eastern CMW. The spatial average VC was 0.4. TE ranged from 1.9×10^9 J m^{-2} (lowest in the CM region) to 2.1×10^9 J m^{-2} (highest in the southwest). Field observations and figure 4 indicate that the southwest region, characterized by lower elevation and drier but denser forest cover, exhibits higher TE due to increased surface radiation absorption and reduced longwave emission. These conditions elevate atmospheric kinetic energy, contributing to local *T* increases. In contrast, the CM region, being cooler, cloudier, and more elevated, exhibited reduced TE, largely due to high albedo from cloud cover and vegetation.

These spatial TE differences reflect physical energy exchange processes consistent with earlier findings by Wondie and Abeje (2022), who reported a negative correlation between outgoing longwave radiation and atmospheric moisture, alongside increases in air *T*. Air *T* and aST followed a similar spatial gradient, with maximum values in the low-lying Abbay Gorge (290 K and 296 K, respectively) and minimum values in the CM (285 K and 286 K, respectively). The average values across CMW were 287 K

Table 5. Decadal variation of the spatial mean values of VC, total evapotranspiration (TE), aST, air *T*, *P*, and IWB at the CMW, Ethiopia.

Year	VC (0–1)	TE (10^9 J m^{-2})	aST (K)	*T* (K)	*P* (mm)	IWB (mm)
1991–2000	0.4	2.0	291.0	287.0	0.70	0.07
2001–2010	0.3	2.0	293.0	288.0	0.70	0.01
2011–2020	0.2	2.1	293.5	288.7	0.65	−0.02
2021–2030	0.3	2.0	294.4	291.5	0.66	−0.05

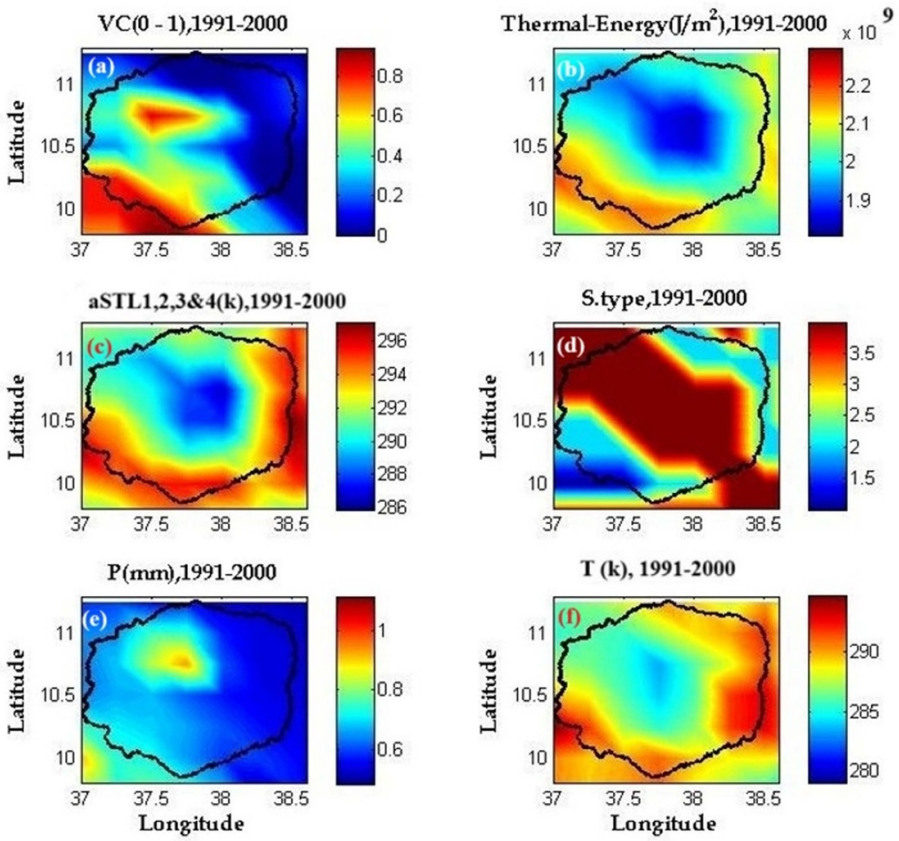


Figure 4. Spatial modeling of the dynamics from 1991 to 2000: VC (panel (a)), TE (panel (b)), aST at $L_1 = 0$ m, $L_2 = 7$ m, $L_3 = 28$ m, $L_4 = 100$ m) (panel (c)), soil type (S. type) (panel (d)), *P* (panel (e)), and air *T* (panel (f)).

(air *T*) and 291 K (aST). Notably, aST exceeded air *T*, which can be attributed to increased solar penetration due to deforestation and the soil's higher capacity to absorb and retain TE. This promotes upward moisture flux and evapotranspiration from soil surfaces. *P* ranged from 0.6 mm in the east to 0.9 mm around CM, with a watershed average of 0.7 mm (table 5, figure 4). In the second decade, VC decreased to a maximum of 0.6, with a spatial average of 0.3 and a 0.1 decline compared to the first decade (figures 4 and 5, panel (a)). The pattern remained spatially consistent, with vegetation loss concentrated in the northern and eastern zones. While TE and *P* values exhibited minimal spatial and temporal variation, both aST and air *T* increased by 2 K and 1 K, respectively. These trends highlight growing thermal stress, likely due to land degradation and vegetation loss, although explicit land-use data would help clarify these drivers.

In the third decade, VC declined further to 0.2, marking a 0.2 reduction relative to the first decade. TE showed a negligible change; however, both aST and air *T* rose by 0.5 K and 0.7 K, respectively. The average IWB shifted from a positive 0.01 mm to −0.02 mm, indicating increasing water stress. Despite only slight increases in *P*, *E*, and *Ro* rose significantly, likely driven by elevated temperature and vegetation loss (table 5, figures 6 and 7). This aligns with hydrological expectations under warming and drying conditions.

In the projected decade (2021–2030), if current land degradation trends continue, IWB is expected to decline to approximately −0.05 mm by 2030. However, afforestation programs underway may mitigate this trend. VC is projected to increase by 0.1 relative to the third decade. Correspondingly, aST

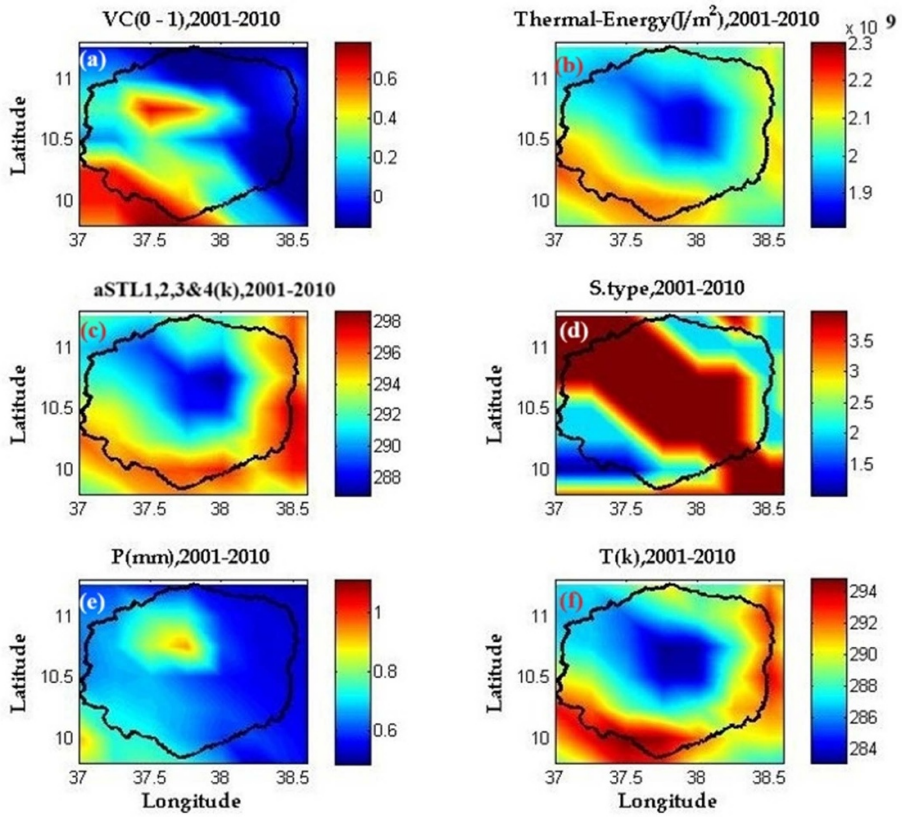


Figure 5. Spatial modeling of the same variables as in figure 4, but for the period 2001–2010.

and air T are expected to rise by 0.9 K and 2.8 K, respectively. The faster increase in air T is attributed to ongoing deforestation and reduced surface moisture, amplifying land–atmosphere energy exchange. These findings align with Demeku *et al* (2022), who observed a decrease in VC and changes in soil color across the CMW, indicative of land degradation. When the WB approaches zero, energy is conserved, and climatic conditions stabilize. A positive WB generally reflects increased P, a negative WB signifies heightened climatic stress manifested as droughts, elevated E, higher wind speeds, and VC loss (figures 3, 5–7). Notably, even when P and E are roughly equal, a positive WB can suppress Ro due to vegetation recovery that enhances infiltration and reduces overland flow.

S. type further informs the hydrological potential and groundwater retention capacity of the CMW. Asmare *et al* (2023) noted that soils derive their characteristics from the parent mineral matter, which can influence soil color and water retention. The dark soils in the central regions and CM are indicative of high organic content and sodium-rich profiles, suggesting favorable GWP (figures 4–7). The bluish and gray soils in the southwest and northeast indicate moisture-saturated conditions. Yellow soils located between the central brown and peripheral gray/bluish zones suggest iron oxidation and leaching processes. The brown tones in central areas imply persistent water saturation throughout the year. Therefore, soil color analysis is a valuable tool for identifying areas prone to waterlogging or nutrient leaching.

3.5. Spatial and decadal analysis of ST and SWV across soil layers L_1–L_4 (1991-2030)

Table 6 and figures 8–11 illustrate the spatial and decadal changes of ST and SWV potential across the CMW, Ethiopia, from 1991 to 2021, with projections through 2030. This analysis is stratified by four soil depth layers (L_1–L_4), enabling detailed vertical and temporal assessments of hydrothermal conditions. During the first decade, the highest SWV potential was recorded at CM in layer L_1, reaching 0.30 m^3 of soil water content per 1 m^3 of soil. Moderate SWV potential of 0.25 m^3 was observed in the northwest of the study area, while the eastern and southwestern regions, particularly around Abbay Gorge, exhibited the lowest SWV potential at 0 m^3. The spatial average SWV potential at L_1 during this period was 0.20 m^3. At layer L_2, the SWV potential peaked at 0.35 m^3 at CM, with a broader geographic coverage compared to L_1. However, the minimum value at L_2 was identical to that at L_1, resulting in a spatial

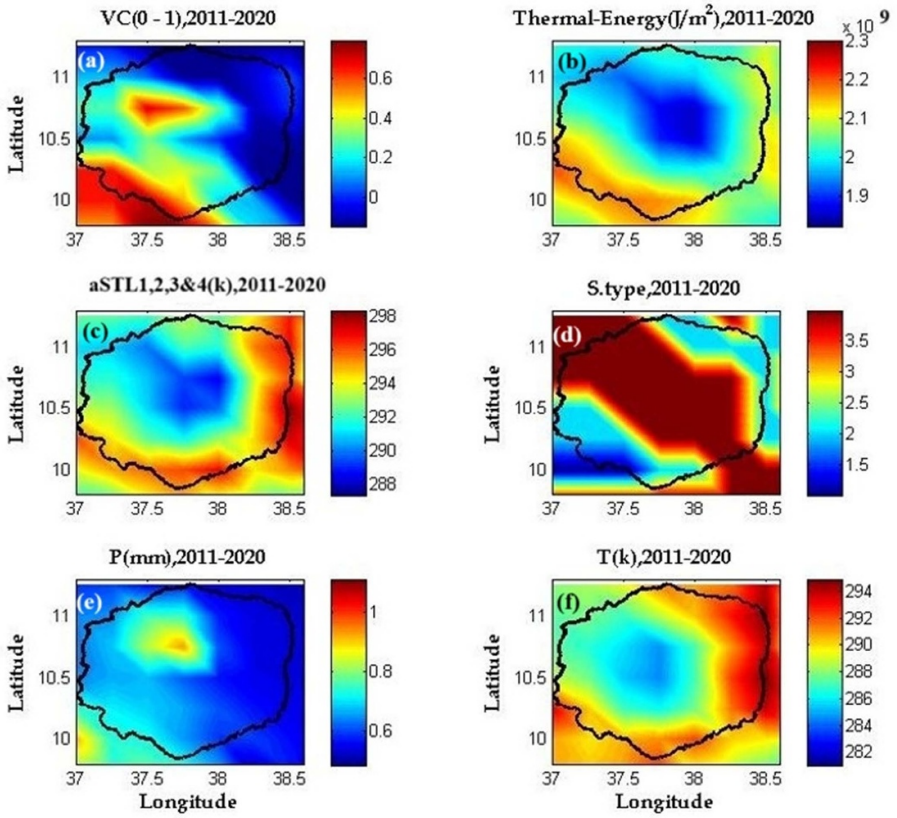


Figure 6. Spatial modeling of the same variables as in figure 4, but for the period 2011–2020.

average SWV of 0.25 m^3. The highest SWV potential at layer L_3 reached 0.37 m^3, excluding the north-east and southern sections of Abbay Gorge, with a mean spatial value of 0.30 m^3. At layer L_4, 97% of the study area demonstrated a water potential of 0.38 m^3, with a maximum value of 0.40 m^3; the spatial average SWV at L_4 was 0.37 m^3, as shown in panels (a), (c), (e), and (g) of figure 8 and table 6.

The maximum SWV potential pattern for the second decade is similar to that of the first decade. The spatial SWV pattern remained broadly similar to the previous decade. Spatial average ST values increased by depth: 287 K (L_1), 290 K (L_2), 292 K (L_3), and 293 K (L_4). The incremental change in ST across layers (ΔST) was most pronounced between L_1 and L_2 (3 K), decreasing gradually toward L_4. Spatial average SWV values for L_1, L_2, L_3, and L_4 were 0.20, 0.26, 0.31, and 0.36 m^3, respectively. The increase in SWV between L_2 and L_3 was 0.05 m^3, but surprisingly, a slight decrease (–0.01 m^3) occurred from L_3 to L_4. ST exhibited a downward trend across layers from the first to the second decade: -1 K at L_1 and L_2, -3 K at L_3, and -2 K at L_4 (figures 8 and 9 panels (b), (d), (f), and (h)). Despite this cooling, SWV remained relatively stable during this period.

In the third decade, SWV continued to decline: spatial averages fell to 0.16 (L_1), 0.22 (L_2), 0.28 (L_3), and 0.34 m^3 (L_4). These represent reductions of 0.04, 0.04, 0.03, and 0.02 m^3, respectively, from the previous decade. In contrast, ST increased across all layers 289 K (L_1), 293 K (L_2), 294 K (L_3), and 294 K (L_4), indicating a warming trend that is likely linked to increased evapotranspiration and declining vegetation cover (figure 10). T increases with decade were more pronounced at shallower depths (L_1, L_2) than deeper layers, consistent with expected surface warming dynamics. Additionally, observed shifts in water table depth suggest increased groundwater depletion during this period.

In fourth-decade projections (2021–2030), if current climate trends persist, the spatial average SWV potential is projected to decline further by 2030, reaching 0.10 m^3 (L_1), 0.18 m^3 (L_2), 0.21 m^3 (L_3), and 0.29 m^3 (L_4). Concurrently, ST is expected to rise to 290 K (L_1), 294 K (L_2), 295 K (L_3), and 294 K (L_4), with minimal spatial variation. The stabilization of ST at deeper layers (L_3 and L_4) suggests thermal equilibrium at depth, while continued reductions in SWV, especially at shallower layers, raise concerns about topsoil desiccation and groundwater depletion. These conditions increase the risk of slope instability and landslides, particularly in elevated terrains (table 6 and figure 11 panels (b), (d), (f), and (h)). Correlation analysis (table 7) highlights a strong negative relationship between VC and average ST ($R^2 = -0.75$, $p < 0.01$), indicating that reduced vegetation leads to surface warming. In contrast, soil

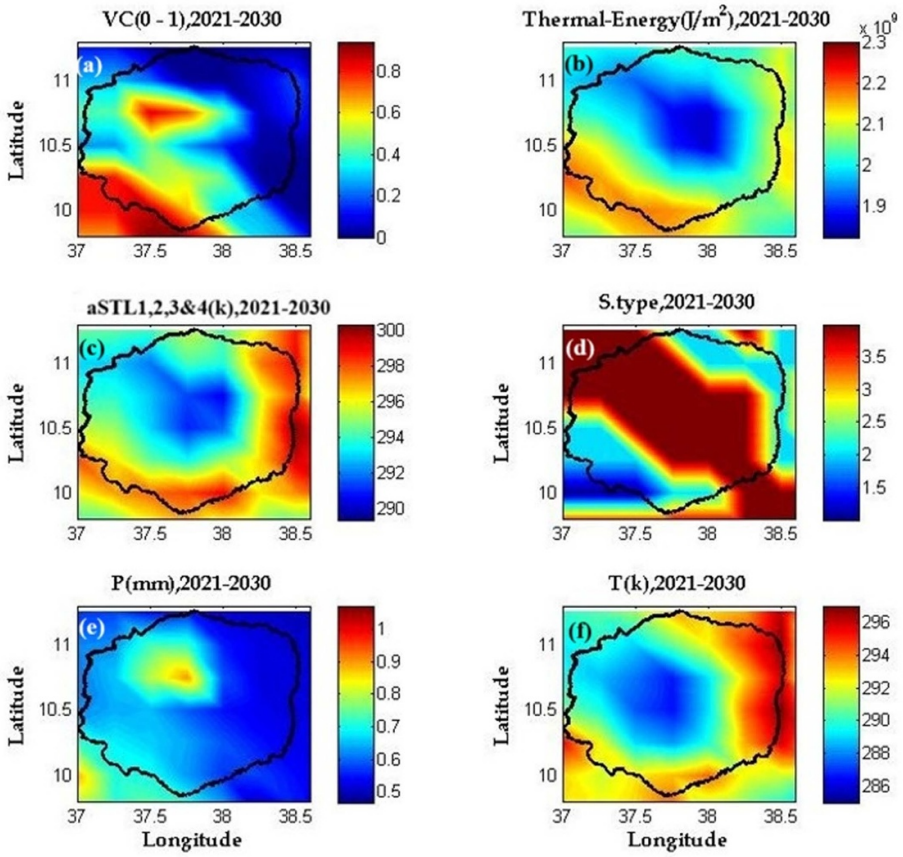


Figure 7. Spatial modeling of the same variables as in figure 4, but showing predicted values for the period 2021–2030 using the ANN model.

Table 6. Decadal variation of ST and volumetric soil water potential (VSWP) at different depths ($L_1 = 0$ m, $L_2 = 7$ m, $L_3 = 28$ m, $L_4 = 100$ m).

Year	Level (L)	ST	VSWP
1991–2000	L1	287	0.20
	L2	290	0.25
	L3	292	0.30
	L4	293	0.37
2001–2010	L1	286	0.20
	L2	289	0.26
	L3	289	0.31
	L4	291	0.36
2011–2020	L1	289	0.16
	L2	293	0.22
	L3	294	0.28
	L4	294	0.34
2021–2030	L1	290	0.10
	L2	294	0.18
	L3	295	0.21
	L4	294	0.29

type (S. type) exhibits a weak and statistically insignificant positive correlation with VC ($R^2 = +0.25$, $p = 0.12$). A strong positive correlation between VC and SWV ($R^2 = +0.81$) confirms that vegetation promotes groundwater recharge and supports a higher water table. These results underscore the critical role of vegetation in microclimate regulation and hydrological sustainability. Accordingly, reversing rising ST and declining SWV will require targeted land restoration, such as afforestation and erosion control.

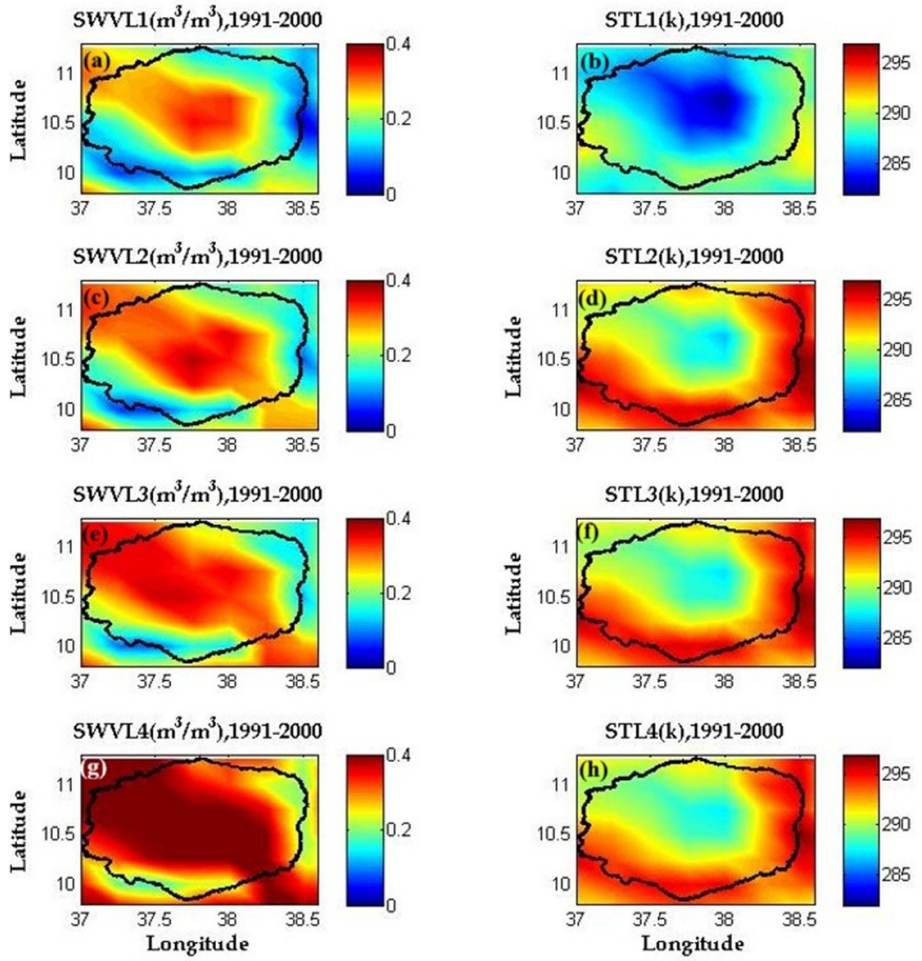


Figure 8. Spatial variation in SWV by depth from L_1 to L_4 (panels (a), (c), (e), and (g)) and ST from L_1 to L_4 (panels (b), (d), (f), and (h)) for the period 1991–2000.

Over the past four decades, groundwater levels in the CMW have declined steadily, especially in highland regions, as shown in figures 8–11. This trend is linked to greater irrigation demand during growing seasons and intensified climate change effects. The resulting land degradation exacerbates water scarcity, increases soil evaporation, and weakens slope stability. While air T exerts a limited direct influence on the WB, P, and E are key drivers (table 3). Soil moisture and groundwater were also assessed at multiple depths: at 0 m, SWV ranged from 5%–30% (avg. 20%); at 7 m, 10%–40% (avg. 25%); at 28 m, 15%–40% (avg. 30%); and at 100 m, 25%–45% (avg. 40%). These vertical gradients reveal that high water potential areas are predominantly located in the western and central parts of the study area. This aligns with conclusions by Mann and Schmidt (2003) and Chapman *et al* (2011), who argue that T changes do not necessarily equate to shifts in WB. In contrast, Taylor and Stefan (2009) suggested shallow groundwater is less sensitive to warming, while Mayer (2012) and others demonstrated strong surface–subsurface interactions. These findings highlight the urgency of integrated land and water resource planning, with VC and riparian ecosystems acting as thermal buffers and hydrological stabilizers. Consistent with Alexander (2006) and Hebert *et al* (2011), this study affirms the value of watershed-scale restoration to build resilience in the face of climate variability.

3.6. The role of Choke Mountain in precipitation formation and its watershed economy valuation

Figure 12 illustrates the spatial variability of U, V, and W winds, along with geopotential height (GPH). Moisture-laden U and V winds originating from the Indian and Atlantic Oceans transport moisture across to the African continent, including Ethiopia (Megbar *et al* 2019). In the absence of topographical barriers, these winds would continue their advection path uninterrupted. However, CM, which peaks at 4094 m, alters this dynamic. As these winds encounter CM, the advection is transformed into orographic convection, generating mountain waves due to vertical wind shear (figure 12, panel (c)). The interaction

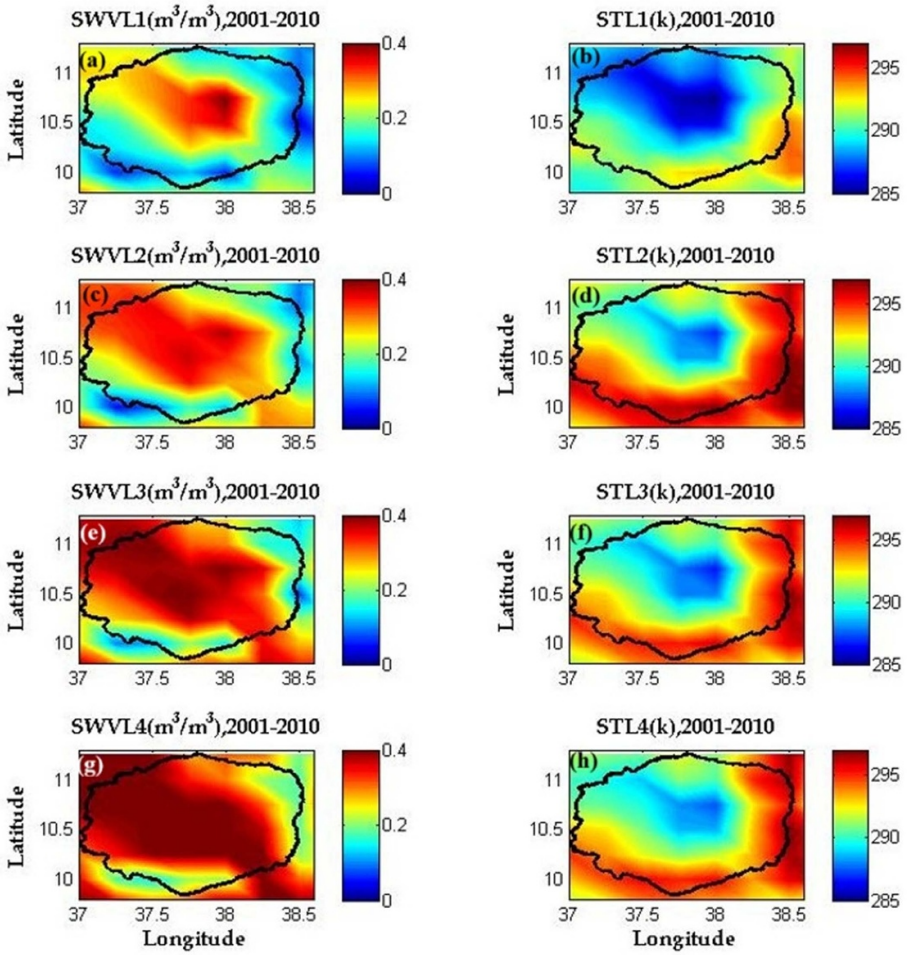


Figure 9. Spatial variation in SWV and ST by depth, similar to figure 8, but for the period 2001–2010.

of these waves with the peak GPH and Coriolis forces causes the winds to curl, facilitating cloud formation and enhancing condensation. This leads to *P* due to gravity-driven deposition of condensed water droplets (figure 13).

Model simulations based on varying GPH (equations (13) and (17)) in figure 13 demonstrate CM's substantial contribution to *P*. *P* sources include both local and large-scale moisture. While local moisture depends on near-surface availability and not altitude, large-scale moisture transport is significantly influenced by topography. When model altitudes are set below 1000 m, *P* is brought only from minimal local moisture (0.05 mm). At 2000 m, a modest increase is observed, corresponding to partial wind obstruction. Significant *P* enhancement occurs at 3000 m and peaks at 4100 m, reaching 1.3 mm (spatial average of 0.75 mm), confirming CM's role in blocking and lifting moisture-laden winds. Radar-based research by Megbar *et al* (2018) further supports this, identifying cloud formation heights between 2500 and 3500 m over the UBNB. In the absence of CM, regions such as East Gojjam, West Gojjam, and parts of Awi would primarily depend on large-scale dynamics for *P*. Thus, CM serves as a critical water-generating feature. The minimal contribution of local moisture to *P* in CM's lower elevations (<1000 m) is evident in figures 13 and 14. Megbar *et al* (2019) noted that local moisture recycling contributes about 20% to total *P* in UBNB. Given CMW's smaller area, its local moisture contribution is even lower.

Temporal *P* trends linked to elevation show decreasing patterns between 2005 and 2011 (figure 14), attributed to the negative phase of the Pacific Decadal Oscillation (PDO) weakening large-scale dynamics (Xianglin *et al* 2023). Peaks from 2014 to 2017 and declines post-2020 also correspond to PDO fluctuations. Similarly, the negative phase of the North Atlantic Oscillation (NAO) from 2000 to 2011 diverted moisture transport away from Africa, further reducing *P* in CMW. These findings are supported by Kim *et al* (2024), validating the interaction between Atlantic and Pacific systems and CMW *P*. Figure 15 shows temporal SWV changes across different soil depths. Surface SWV remains stable, but deeper layers (28 m and 100 m) show a gradual decline between 1993 and 2020. A slight rebound at 7 m and 28 m

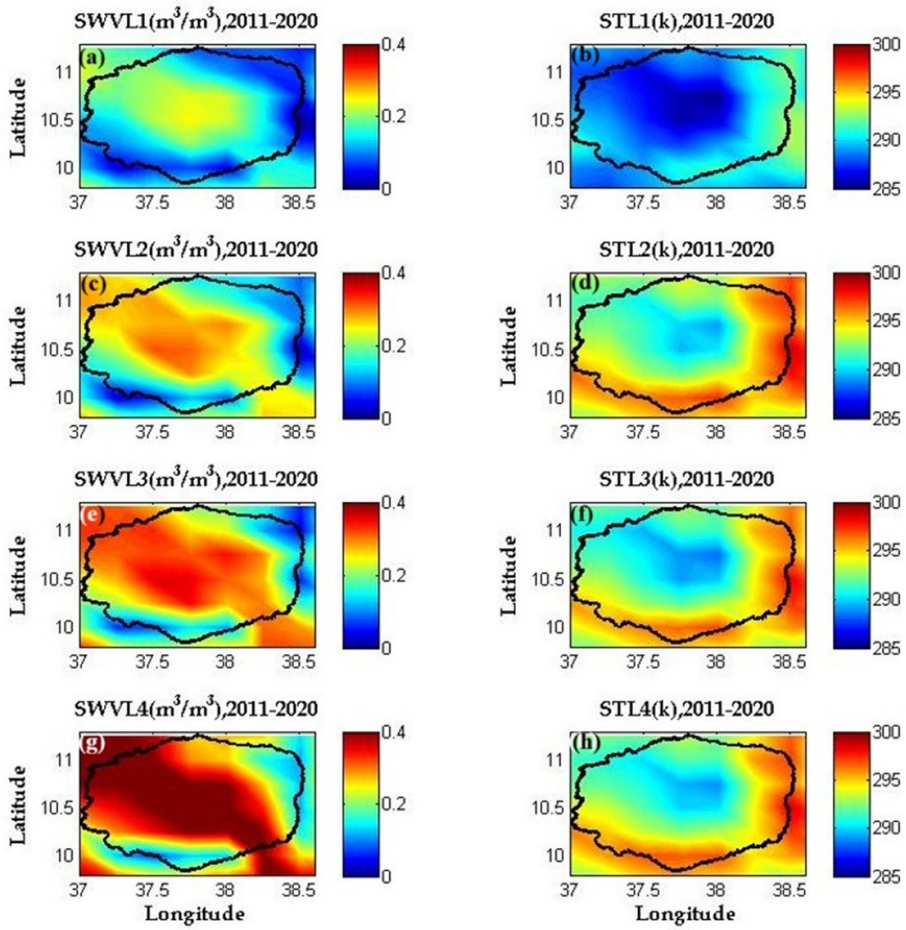


Figure 10. Spatial variation in SWV and ST by depth, similar to figure 8, but for the period 2011–2020.

occurs from 2020 to 2024, while 100 m continues to decline. This indicates a slow retreat of the water table. Using 35 year average dry season (DJF) data, mean groundwater storage was 0.05 m per 1 m^3 of soil (figure 16, panel (a)). Annual averages, incorporating all seasons and storage, show 0.50 m per 1 m^3 soil (figure 16, panel (c)). Cumulative annual P (see equation (13)) totals 9.50 m at the Earth's surface, while the GWP in depth averages 13.5 m per 100 m soil depth area (panel (d) in figure 16). Based on the spatial extent of the CMW, which is 27 722.25 km^2 (2.772 $\times$ 10^{10} m^2), the net annual water potential from the surface and ground sources is approximately 1.386 $\times$ 10^{10} m^3, while total precipitation (TP), excluding losses, is 2.634 $\times$ 10^{11} m^3. Economic analysis estimates CMW's atmospheric and groundwater outflow to be worth \$70 billion annually. Groundwater sources alone contribute around 1.39 $\times$ 10^9 m^3 (\$4.9 billion), and the total water value, including TP and GWPS, is \$133.4 billion annually (table 8).

Figure 16 also shows that GWP is greatest near the CM peak (depth of 20 m per unit area), with the lowest values in eastern and southwestern zones, including Abbay Gorge. Spatial average groundwater depth is 13 m per 100 m of soil. Sisay *et al* (2024) predict groundwater decreases of 3.24% and 6.24% in the Katar subbasin and 7.43% and 0.29% in the Meki subbasin by the 2040s and 2070s, respectively, under RCP4.5. These findings align with our study and underscore the importance of hydro-economic models (Baker *et al* 2021). The CMW groundwater valuation model links water supply and demand through a system of parameterized economic trade-offs and marginal benefits, fostering better water governance. Finally, P flux analysis suggests that drought years (2029, 2040, 2058) and flood years (2045, 2059, 2061, 2069) (figure 17) are identified. Effective planning, awareness, and climate adaptation strategies are vital. If land-use and mitigation strategies remain unchanged, variability in P, E, and Ro will intensify, although the magnitude of fluxes may vary over time. This highlights the need for proactive and adaptive watershed and land management strategies.

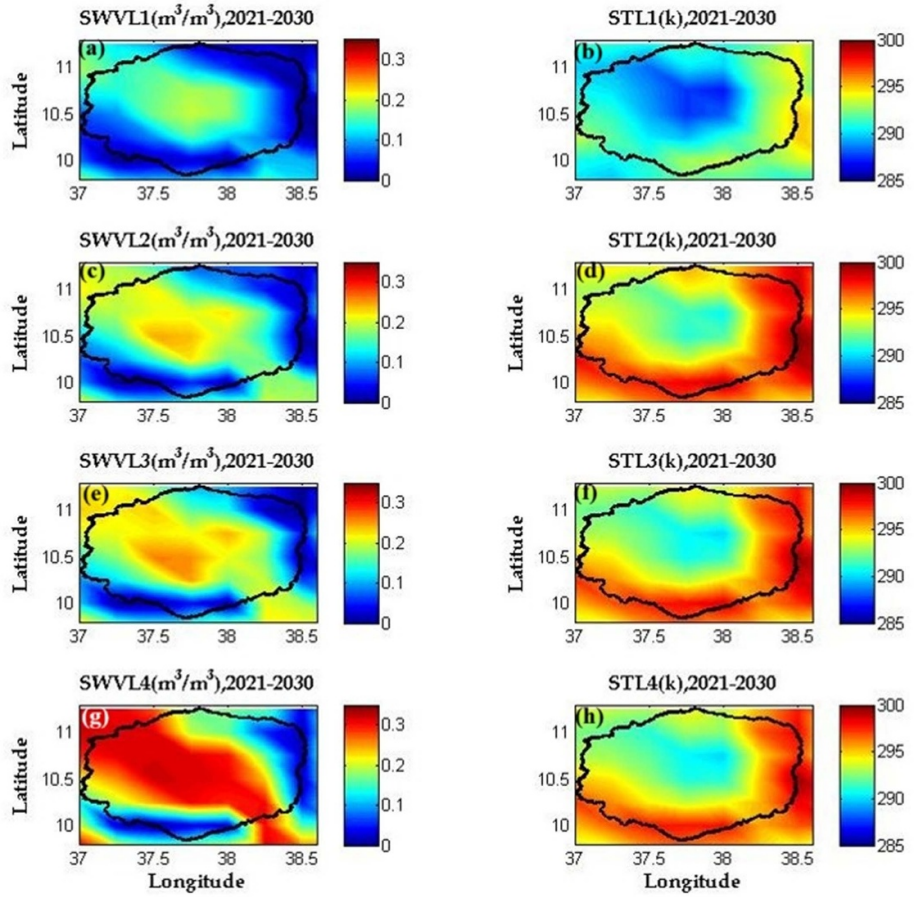


Figure 11. Spatial variation in SWV and ST by depth, similar to figure 8, but for the period 2021–2030.

Table 7. R^2 and significance level (P-value) between VC and S. Type, aST), and average volumetric soil water (aVSW) in the study area from 1990 to 2024.

VC	R^2	P
S. type	0.25	0.12
aST	−0.75	0.01
aVSW	0.81	0.01

4. Discussion

ERA5 shows the strongest agreement with *in-situ P* over the CMW, combining high R^2 with near-zero BR and low MAE, indicating reliable representation of P variability and magnitude. CHIRPS and CRU perform moderately but exhibit systematic overestimation, while the RCM shows weak correlation despite low RMSE, likely reflecting limited skill in capturing temporal variability. The superior performance of ERA5 is consistent with earlier findings in the UBNB (Abera *et al* 2017, Megbar *et al* 2019), supporting its suitability for regional WB and hydroclimatic assessments.

The comparison of the OWB and IWB models over the CMW shows that incorporating additional hydrological processes substantially improves WB realism. The OWB consistently yields negative, weakly declining balances, indicating limited ability to capture hydrological variability and mass continuity. In contrast, the IWB exhibits both positive and negative oscillations, better reflecting seasonal and interannual dynamics (see table 3, figures 2 and 3). Similar long-term tendencies have been reported in Ethiopian watersheds using SWAT and other models (Bedasa and Yilma 2024, Tadess and Jothimani 2024, Tadie 2024), though the IWB captures additional spatial variability linked to topography and land cover. Enhanced *P*, *Ro*, and storage around CM highlight strong orographic and terrain controls, consistent with climate sensitivity patterns observed in highland regions (Mulvaney *et al* 2022).

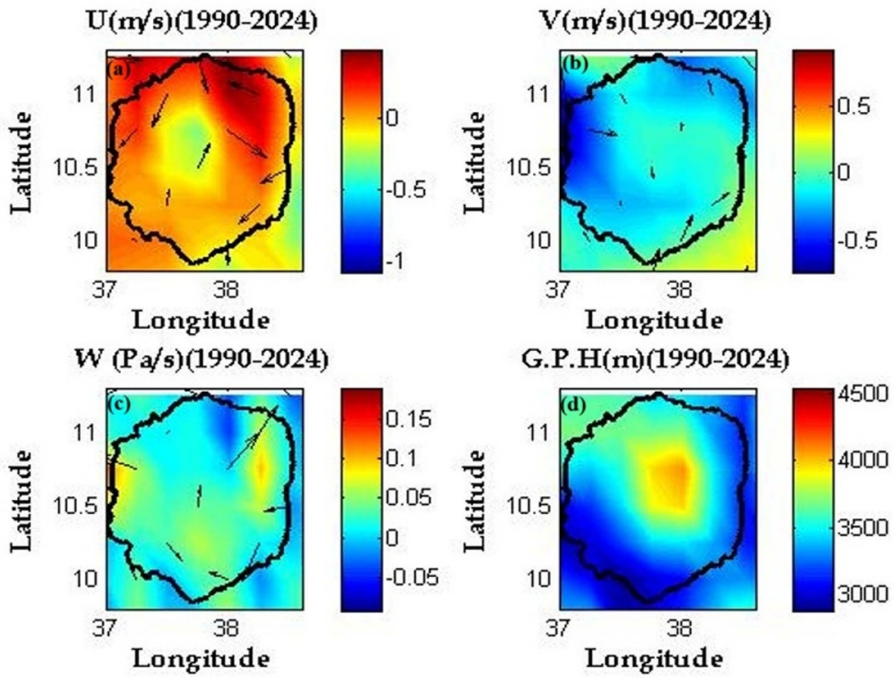


Figure 12. Spatial variation of *U, V*, vertical wind shear (*W*), and GPH from 1990 to 2024. Black arrows indicate wind direction, while the color shading represents wind magnitude.

Strong positive associations of IWB with *P* and storage, and negative links with evapotranspiration and *Ro*, underscore the dominance of surface processes and support the use of refined, locally tailored models for sustainable water resource management.

The decadal and spatial analysis of biophysical variables across the CMW reveals strong coupling between VC, TE, *T, P*, and soil properties (table 5, figures 4–7). High VC and lower TE over CM contrast with warmer, energy-rich lowland areas, reflecting elevation- and land–cover–controlled energy exchange, consistent with Wondie and Abeje (2022). The sustained decline in vegetation since the 1990s, accompanied by rising air and soil *T*, indicates increasing thermal and water stress, in line with land degradation trends reported by Demeku *et al* (2022). Higher ST than air *T* suggests enhanced solar absorption linked to deforestation, intensifying evapotranspiration. Spatial soil color patterns further indicate variable moisture retention and GWP, consistent with soil genesis interpretations by Asmare *et al* (2023), highlighting increasing hydroclimatic vulnerability in the CMW.

The decadal evolution of ST and SWV across the CMW reveals increasing hydrothermal stress with important implications for groundwater security and slope stability. Early decades show high SWV at depth around CM, indicating favorable recharge conditions, while persistent low SWV in the Abbay Gorge reflects topographic and land-use constraints. From the third decade onward, rising ST, especially in shallow layers, coincides with declining SWV, signaling progressive soil desiccation, groundwater depletion, and elevated landslide risk in steep terrains (tables 6 and 7, figures 8–11). These dynamics align with surface subsurface coupling reported by Mayer (2012) and Chapman *et al* (2011), and contrast with studies suggesting weak groundwater sensitivity to warming (Taylor and Stefan 2009). Strong positive VC and SWV and negative VC and ST relationships confirm vegetation's buffering role, supporting afforestation and watershed restoration as priority climate-adaptation strategies, consistent with Terzer Alexander (2006).

The spatial and temporal analysis of winds, *P*, and groundwater in the CMW highlights CM as a critical regulator of regional hydroclimate. Moisture-laden *U* and *V* winds are forced into orographic convection by CM, generating mountain waves that enhance cloud formation and *P* (table 8, figures 12–17). This study is consistent with radar observations in the UBNB (Megbar *et al* 2018, 2019). *P* trends reflect modulation by Pacific and Atlantic oscillations (PDO and NAO), corroborating the role of large-scale climate drivers in Ethiopia (Ren *et al* 2023). SWV and groundwater analyses reveal declining water storage at depth, particularly in eastern and southwestern zones, echoing predictions by Sisay *et al* (2024). These findings underscore the importance of vegetation, topography, and subsurface storage for

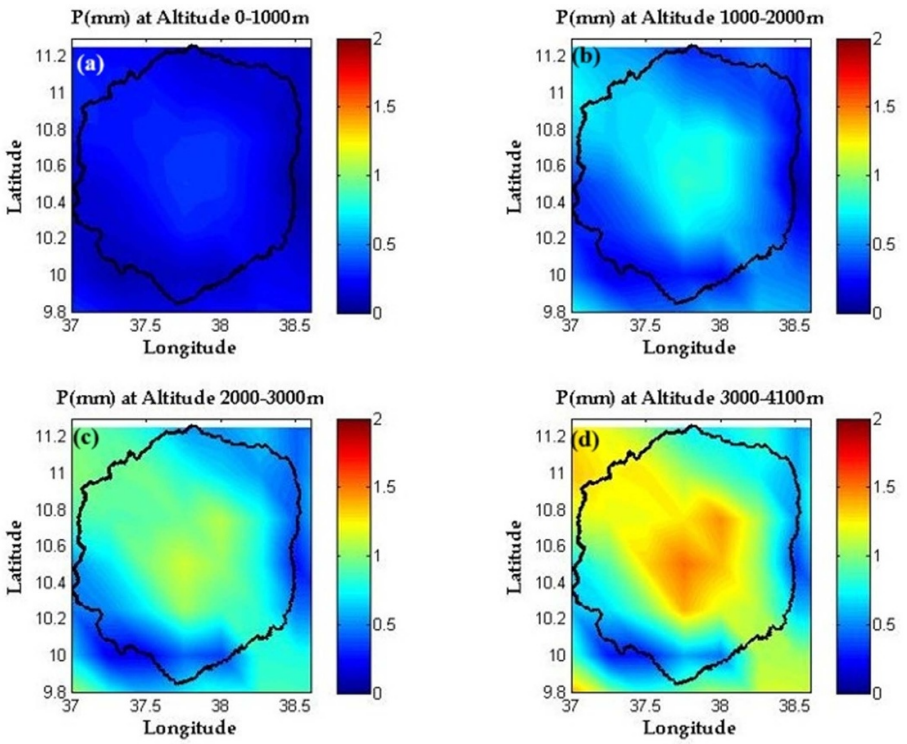


Figure 13. Model results illustrating the role of mountains in the mean total *P* contribution and its spatial variability across different altitude ranges (0–1000 m, 1000–2000 m, 2000–3000 m, and 3000–4100 m) in the CMW from 1990 to 2024.

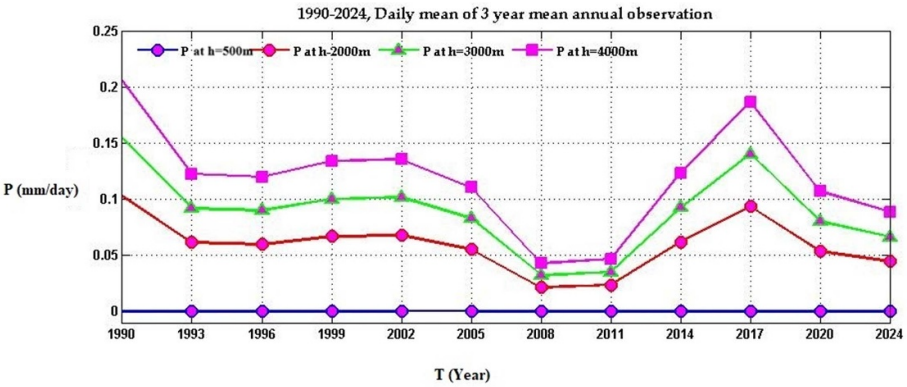


Figure 14. Model results showing the role of mountains in the mean total *P* contribution and its temporal variability at different altitudes (0 m, 2000 m, 3000 m, and 4000 m) in the CMW from 1990 to 2024, presented at three-year intervals.

water security, while the hydro-economic valuation emphasizes the need for integrated watershed management and adaptive land-use strategies to mitigate future droughts, floods, and groundwater depletion.

Temporal *P* in the CMW shows clear links to large-scale climate variability. Between 2005 and 2011, *P* decreased across elevations, coinciding with the negative phase of the PDO that weakened regional moisture transport (Ren *et al* 2023). Peaks from 2014 to 2017 and declines after 2020 reflect further PDO fluctuations. Similarly, the negative phase of the NAO from 2000 to 2011 diverted moisture away from Africa, reducing rainfall in the CMW. These findings align with Kim *et al* (2024), confirming the interaction of Atlantic and Pacific systems in modulating highland *P*.

SWV analysis reveals stable surface moisture but declining deeper layers (28 m and 100 m) from 1993 to 2020, indicating slow groundwater depletion. Slight recovery at 7 m and 28 m after 2020 contrasts with a continuous 100 m decline, suggesting retreat of the deep-water table. Groundwater storage averages 0.05 m per m^3 of soil in the dry season and 0.50 m annually, with cumulative annual *P* of 9.50 m and mean groundwater depth of 13.5 m (figure 16). Scaling to 27 722.25 km^2 yields a net annual

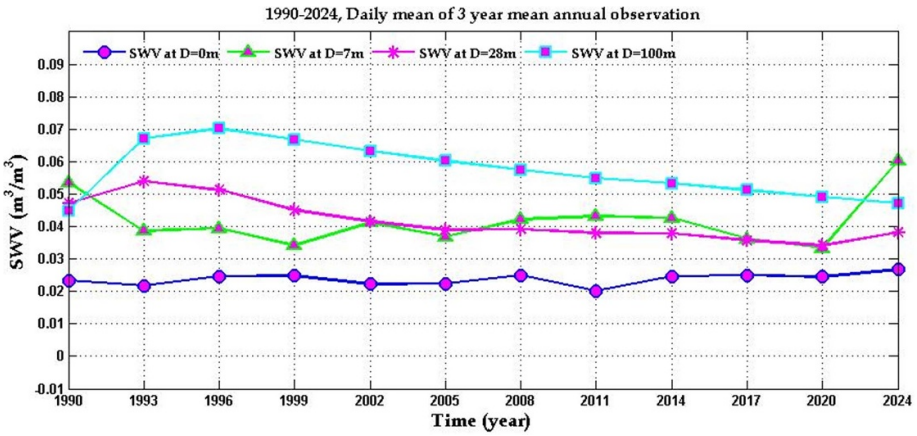


Figure 15. Temporal variation of SWV at depths of 0 m, 7 m, 28 m, and 100 m in the CMW from 1990 to 2024.

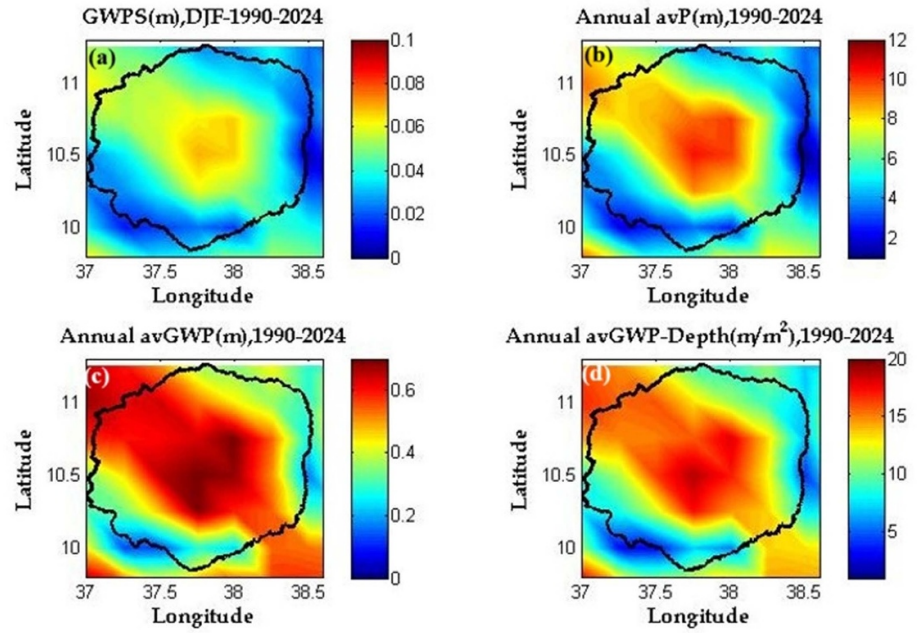


Figure 16. The average value of the December, January, and February the mean value of 35 years SWV data ground water potential only ground sources without storage (GWPS) in m at depth 0–1 m panel (a), total atmospheric annual average value of 35 years data of P in m plotted at panel (b), annual average 35 years data value of GWP at 0–1 m represented at panel c, and the annual ground water depth (H) from the integral value of GWP from $L = 0$ to $L = 100$ m depth per unit area of the CMW spatial distribution represented at panel (d) from 1990–2024.

water potential of 1.386×10^{10} m^3 and total potential water (TP) of 2.634×10^{11} m^3, highlighting substantial water resources.

Economic valuation estimates annual atmospheric and groundwater outflows at $70 billion, with groundwater alone contributing $4.9 billion. Including TP and GWP storage, the total water value reaches $133.4 billion (table 8). These results underscore the importance of integrating hydrological modeling with economic assessment for sustainable water management in water-scarce regions (Birol *et al* 2006, 2010, Al-Juaidi *et al* 2014, Ortiz-Partida *et al* 2023).

Overall, the CMW exhibits high sensitivity to climatic oscillations and groundwater stress, while its water resources hold significant economic and ecological value. Integrating hydroclimatic data with economic metrics provides a robust basis for informed resource allocation and sustainable watershed management.

Table 8. Surface and groundwater water resource indicators and their economic valuations at the CMW: surface water potential from the ground and atmosphere (SWPGA) that flows outside the study area; groundwater resources excluding atmospheric water (GWR); atmospheric water resources as a portion of total precipitation (TP); water resources of the atmosphere generated due to the mountain and the groundwater economy valuation (WRAGEV); groundwater resources economic valuation (GWREV); atmospheric water economic valuation (TPEV); and surface water resources from the ground and atmosphere economic valuation flowing outside the CMW per year (SWRGAEV).

Parameters	Average value in m	Total value in m^3 per CMW
TP	9.5	2.634×10^{11}
GWR	0.05	1.390×10^{9}
SWPGA	0.5	1.386×10^{10}
GWR + TP	10	2.647×10^{11}

WRAGEV = CCF * MAB = $133.4 billion
GWREV = CCF * MAB = $4.9 billion
TP = CCF * MAB = $128.5 billion
SWRGAEV = CCF * MAB = $70.0 billion

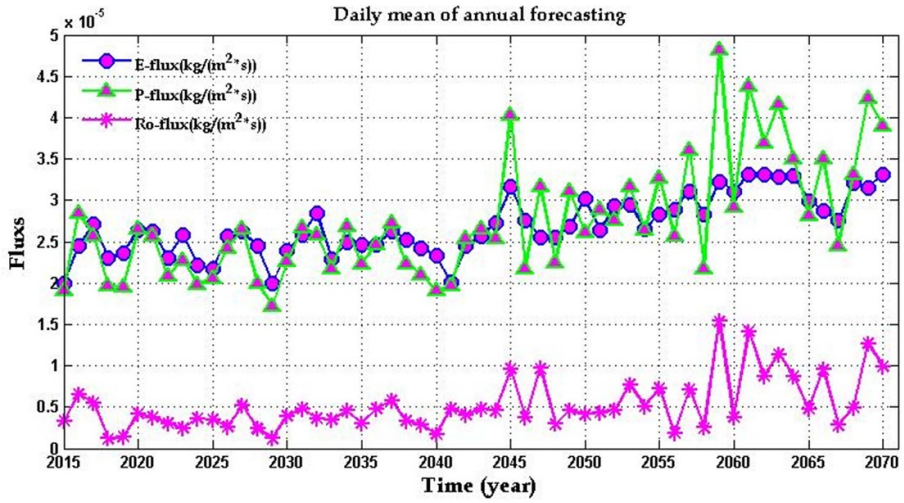


Figure 17. Predicted daily means of annual *E*, *P*, and *Ro* flux from 2015 to 2070. All three fluxes show an increasing trend, indicating heightened variability due to climate change. While extreme event frequency is expected to rise, the average *P* may remain constant.

5. Conclusions

This study developed and implemented an altitude-sensitive water resources estimation model for economic valuation in the CMW. The enhanced WB model demonstrated superior accuracy compared to the original version, as evidenced by assessments of soil and atmospheric moisture dynamics, vegetation cover, and the spatiotemporal variability of hydroclimatic conditions. A model was also developed to quantify the influence of CM topography on *P* formation, incorporating altitude variation, SWV estimates, and wind dynamics. The net annual water from CMW to downstream countries is estimated at 13.86 billion m^3, with an economic value of $70.0 billion. Without proper watershed management, CMW could lose up to 1.386 billion m^3 of water annually, resulting in significant economic losses. To guide future policy and sustainable resource management, four potential scenarios for the CMW's hydrological future are proposed:

1. Environmental Safeguarding: A proactive approach involving researchers, policymakers, and communities could maintain or enhance CMW water resources. This scenario depends heavily on reforestation, erosion control, and land restoration efforts.

2. Typical Environmental Management: If existing practices persist, the average groundwater depth may fall below 10 m per 100 m of soil, correlating with increased variability in *P*, *E*, and *Ro*. This result suggests increased vulnerability in the hydrological system, especially in the context of climate change.
3. The partial degradation of the CMW: Moderate land degradation could reduce soil moisture by up to 50%, increasing the risk of landslides, erosion, and reduced mountain elevation (up to 5 m). This would also reduce daily *P* by 0.02 mm per year, with negative ecological and hydrological implications.
4. Significant Degradation: Significant environmental degradation could lead to a 10 m reduction in mountain elevation due to landslides. This would result in an annual loss of 1.386 billion m^3 from reservoirs such as the GERD and a 0.04 mm per year decrease in *P*, translating into an estimated $0.5 billion annually in economic loss.

Future studies should couple the IWB framework with climate change scenarios and land-use projections to support long-term adaptation strategies and transboundary water cooperation in the Nile Basin.

Data availability statement

The ERA5 data can be accessed at www.cds.climate.copernicus.eu (Hersbach *et al* 2020), while the CHIRPS data is located at https://data.chc.ucsb.edu/products/CHIRPS/v3.0/README-CHIRPSv3.0.txt (Ahmed *et al* 2024). RCM data can be found at https://cds.climate.copernicus.eu/datasets/projections-cordex-domains-single-levels?tab=overview (Copernicus Climate Change Service, Climate Data Store 2019, Yersaw and Chane 2024), and the CRU data is accessible at https://crudata.uea.ac.uk/cru/data/hrg/ (Harris *et al* 2020, Morsy *et al* 2022), and the *in situ* data provided by the National Meteorology Institute of Ethiopia are publicly accessible. The Reid AR identification algorithm (Reid *et al* 2020) is hosted on GitHub-Zenodo at https://doi.org/10.5281/zenodo.3748269.

Ethical statement

This study did not involve human participants or animals. It is based solely on hydrological and environmental data; therefore, ethical approval was not required.

Author contributions

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